

Transposable Elements and Related Sequences

Both prokaryotes and eukaryotes have stretches of DNA that can move from one location to another within the genome. These stretches are known as *transposable genetic elements*, or simply **transposable elements**. During the process called *transposition*, a transposable element moves from one site in a cell's DNA to a different target site by a type of recombination process. Transposable elements are sometimes called “jumping genes,” but actually they never entirely detach from the cell's DNA. Instead, the original and new DNA sites are brought very close together by enzymes and other proteins that bend the DNA. Surprisingly, about 75% of human repetitive DNA (44% of the entire human genome) is made up of transposable elements and sequences related to them.

The first evidence for wandering DNA segments came from American geneticist Barbara McClintock's breeding experiments with calico corn (maize) in the 1940s and 1950s (**Figure 21.7**). Tracking corn plants through many generations, McClintock analyzed changes in the color of corn kernels. The patterns she saw led her to propose that there were genetic elements capable of moving from other locations in the genome into the genes for kernel color, disrupting the genes and changing the kernel color. McClintock's hypothesis provoked great interest among her maize colleagues, but most other scientists thought the phenomenon she had observed might occur only in maize. Her careful work and insightful ideas were finally validated many years later when transposable elements were found in bacteria. In 1983, at the age of 81, McClintock received the Nobel Prize for her pioneering research.

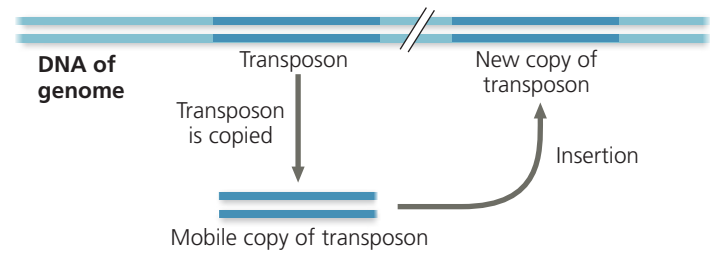
Movement of Transposons and Retrotransposons

Eukaryotic transposable elements are of two types. The first type, **transposons**, move within a genome by means of a DNA intermediate. Transposons can move by a “cut-and-paste”



◀ **Figure 21.7** The effect of transposable elements on corn kernel color. Barbara McClintock first proposed the idea of mobile genetic elements after observing variegations in the color of the kernels on a corn cob (top right).

▼ **Figure 21.8** Transposon movement. Movement of transposons by either the copy-and-paste mechanism (shown here) or the cut-and-paste mechanism involves a double-stranded DNA intermediate that is inserted into the genome.



VISUAL SKILLS How would this figure differ if it showed the cut-and-paste mechanism?

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mechanism, which removes the element from the original site, or by a “copy-and-paste” mechanism, which leaves a copy behind (**Figure 21.8**). Both mechanisms require an enzyme called *transposase*, which is generally encoded by the transposon.

Most transposable elements in eukaryotic genomes are of the second type, **retrotransposons**, which move by means of an RNA intermediate that is a transcript of the retrotransposon DNA. Thus, retrotransposons always leave a copy at the original site during transposition (**Figure 21.9**). To insert at another site, the RNA intermediate is first converted back to DNA by reverse transcriptase, an enzyme encoded by the retrotransposon. (Reverse transcriptase is also encoded by retroviruses, as you learned in Concept 19.2. In fact, retroviruses may have evolved from retrotransposons, or vice versa.) Another cellular enzyme catalyzes insertion of the reverse-transcribed DNA at a new site.

▼ **Figure 21.9** Retrotransposon movement. Movement begins with synthesis of a single-stranded RNA intermediate. The remaining steps are essentially identical to part of the retrovirus replicative cycle (see Figure 19.8).

